

Kuiyi (Cory) Gao

Chapel Hill, NC | kuiyigao@unc.edu | kuiyigao.github.io | github.com/KuiyiGao | [Google Scholar](#) | [LinkedIn](#)

Education

University of North Carolina at Chapel Hill

Aug 2026 – May 2027

B.A. in Computer Science (transfer); exchange student Jan – May 2026

(expected)

- Selected coursework: COMP 550 Natural Language Processing, COMP 562 Machine Learning, COMP 586 Advanced NLP (research project), COMP 380 Technology, Ethics & Culture; Fall 2026: COMP 211 Systems Fundamentals, COMP 301 Foundations of Programming.

The Chinese University of Hong Kong

Sep 2023 – May 2026

B.Sc. program in Computer Science, Faculty of Engineering ELITE Stream (transferred to UNC)

- Cumulative GPA 3.44/4.00 (as of the April 2026 transcript).
- Selected coursework: ELEG 5491 Introduction to Deep Learning (graduate), ESTR 3108 Artificial Intelligence (ELITE), Applied Deep Learning, Operating Systems, Probability, Linear Algebra, Database Design.

Publications

- [1] Wenxuan Wang, Xiaoyuan Liu, **Kuiyi Gao**, Jen-tse Huang, Youliang Yuan, Pinjia He, Shuai Wang, Zhaopeng Tu. *Can't See the Forest for the Trees: Benchmarking Multimodal Safety Awareness for Multimodal LLMs*. **ACL 2025 (Main Conference)**. aclanthology.org/2025.acl-long.832
- [2] Wenxuan Wang, **Kuiyi Gao**, Youliang Yuan, Jen-tse Huang, Qiuqi Liu, Shuai Wang, Wenxiang Jiao, Zhaopeng Tu. *Chain-of-Jailbreak Attack for Image Generation Models via Step by Step Editing*. **Findings of ACL 2025**. aclanthology.org/2025.findings-acl.571
- [3] **Kuiyi Gao*** et al. *RealGUINoise: A Dynamic Cross-Platform Benchmark for GUI Agent Robustness under Real-World Interface Noise*. Under review, 2026 (*co-first author).

Research Experience

GUI-Agent Robustness under Interface Noise — CUHK, advised by Prof.

Hong Kong & remote

Michael R. Lyu

Sep 2025 – present

- Co-first author of a dynamic, cross-platform (web / desktop / mobile) benchmark: 42 interface-noise types, 7 agent frameworks evaluated in live executable environments against task-level rubrics; noise redirects trajectories and raises unsafe-action rates, not only failure rates.
- Designed the evaluation metric suite (pass@k, relative degradation, action-difference rate, unsafe-action rate, correct-prefix length) and the mock-interface harness for computer-use, browser-use and self-operating agents.

Agent Skill Firewall — MBZUAI Undergraduate Research Internship Program

Abu Dhabi & Beijing

(UGRIP), Dr. Steve Liu

(hybrid)

Jun – Jul 2026

- Built the runtime egress firewall for agent skills in a four-student team: static pre-scan, then intent–action–result verification through an HTTP proxy with canaries; PASS / DEFER / BLOCK decisions with an evidence record per verdict; TUI and web console; Docker deployment; 23 passing tests with CI on Linux and macOS.
- On our own 112-skill corpus (76 malicious / 36 benign): recall 0.987, 0 false positives; matches a static-plus-LLM scanner's recall (McNemar $p = 1$) at a lower false-positive rate ($p = 0.016$).
- Continuing the work as a measurement study of runtime-detector recall (execution $\times$ observability $\times$ detection), with pre-registered lab experiments; manuscript in preparation.

Chain-of-Jailbreak: Editing-Step Attacks on Image-Generation Models —

Hong Kong

Tencent AI Lab & CUHK; Prof. Michael R. Lyu, Wenxuan Wang, Zhaopeng Tu

Jun 2024 – Feb 2025

- Designed the attack methodology (step-by-step editing that decomposes a harmful query into benign edits); built part of the dataset; ran most attack and defense experiments; wrote the paper and led the rebuttal. Second author, Findings of ACL 2025 [2].

MMSafeAware: Multimodal Safety Awareness in MLLMs — *Tencent AI Lab;* *Hong Kong & Shenzhen*
Zhaopeng Tu, Wenxuan Wang *Dec 2023 – Oct 2024*

- Designed the evaluation methodology, built the benchmark (1.5k image–prompt pairs across 29 safety scenarios, covering both unsafe and over-sensitive responses) and ran the vision-centric reasoning fine-tuning experiments. Third author, ACL 2025 main conference [1].

Memory × Persona Safety — a pre-registered research program *Independent*
 Jul 2026 – present

- Designed and pre-registered five studies (steering dose law, retrieval-induced forgetting, memory confidence without metadata, memory-mediated sycophancy drift, deletion-influence residual audit) with a shared experiment harness (28 smoke tests); proposal stage.

Kernel-Based Novelty Scores for Text — *CUHK summer research, Prof. Farzan* *Hong Kong*
Farnia *May – Aug 2025*

- Explored extending the Kernel Entropic Novelty (KEN) score to text (whitening for isotropy, a differentiable Nyström approximation) and a personality-conditioned multi-objective alignment formulation; proposal stage.

Selected Projects

- **MemSlot saliency + DeepSeek-OCR compression for long contracts** (UNC COMP 586, 2026): frozen RoBERTa with 32 learnable memory slots learns question-agnostic saliency; salient spans are rendered and compressed through DeepSeek-OCR; CUAD metrics with ten control arms. [\[code\]](#)
- **Course AI teaching assistant** (CUHK GEWS2011, 2025): Flask web app whose two-stage LLM pipeline drafts an outline under the course’s academic-integrity guideline before answering — guidance rather than answers; invitation-code accounts, admin console, Render deployment.
- **Objectivity of Imagist poetry measured with text-to-image models** (CUHK ENGE3940, 2025): 20 images per poem from Flux.1-dev and SD3, CLIP-embedding variance as the statistic — Imagism 0.12 vs. Romanticism 0.28 (Flux, $p = 0.0499$).
- **Movability segmentation for embodied safety** (CUHK ELEG 5491, 2025): DeepLabV3 with a custom ASPP head on COCO-Stuff relabelled into four movability levels; mIoU 0.638 at 58 FPS. [\[code\]](#)

Honors & Awards

- CUHK Faculty of Engineering ELITE Stream; Professor Joseph Lau Scholarship of Innovation; CUHK College Entrance Scholarship; CUHK exchange scholarship (UNC, Spring 2026).
- Gaokao (national college entrance examination): top 2% in Beijing, 2023.

Skills

- **Languages:** Python, C, Java, JavaScript/HTML/CSS, $\text{\LaTeX}$. **Frameworks & tools:** PyTorch, Hugging Face Transformers, Flask, Docker, GitHub Actions CI, Playwright / GUI-agent harnesses, LLM APIs (OpenAI, Anthropic, DeepSeek).
- **Natural languages:** Mandarin (native), English (TOEFL iBT 110, Jan 2025), Cantonese (intermediate).